CITY IoT SHIELD

CYBERSECURITY TRACK · HACKATHON SCREENING PROPOSAL

Tagline: *Securing Today's Connected Devices. Protecting Tomorrow's Smart Cities.*

1. Problem Statement

- **Expanding attack surface:** Smart cities increasingly rely on connected traffic systems, CCTV cameras, utility meters and environmental sensors, creating more entry points for cyberattacks.
- **Insecure IoT devices:** Weak credentials, outdated firmware and poorly protected connections can allow attackers to take control of devices or move deeper into city networks.
- **Threats to essential services:** A successful attack could disrupt traffic management, electricity monitoring, water services and other critical public operations.
- **Slow threat detection:** Abnormal device behaviour can remain unnoticed, giving attackers time to cause wider damage before teams can respond.
- **Fragmented visibility:** Devices from multiple manufacturers are often managed through separate systems, making it difficult to maintain a unified view of assets, vulnerabilities and incidents.
- **Privacy exposure:** Compromised devices may reveal residents' personal data, surveillance footage or other sensitive information.
- **Growing systemic risk:** As cities adopt AI-enabled infrastructure, autonomous transport and smart grids, one compromised device could affect interconnected services unless systems are securely segmented.

Why it matters: An estimated 16.6 billion IoT devices were connected worldwide by the end of 2023, underscoring the urgent need to secure the infrastructure on which modern cities increasingly depend.

2. Proposed Solution

City IoT Shield is a proposed cybersecurity platform that gives smart-city teams a unified view of connected devices, identifies suspicious behaviour and supports faster, safer incident response.

Core capabilities:

- **Device discovery and monitoring:** Identify, register and continuously track connected city devices.
- **AI-assisted anomaly detection:** Flag unusual traffic patterns or behaviour that may indicate compromise.
- **Vulnerability alerts:** Highlight devices affected by known weaknesses, weak configurations or outdated firmware.
- **Risk-based dashboard:** Rank alerts by severity, exposure and potential impact on city services.
- **Guided response:** Recommend investigation, isolation, patching and recovery actions for authorised teams.

The objective is practical risk reduction: make smart-city infrastructure more difficult to compromise, help teams detect attacks sooner and support faster recovery—without claiming that any system can be completely immune to cyber threats.

3. Technology Stack

Core Technology Stack

Python — Powers the platform's security logic, data processing and threat-analysis workflows.

FastAPI — Delivers fast, secure backend services and APIs for data exchange.

PostgreSQL — Maintains device inventories, security events, alerts and risk records.

Streamlit — Provides an interactive dashboard for monitoring devices, alerts and risk levels.

Scikit-learn — Enables optional machine-learning models for anomaly detection and risk scoring.

MQTT — Simulates lightweight, real-time communication between connected IoT devices.

This stack is intentionally lightweight, open-source and suitable for rapid prototyping. At the screening stage, it demonstrates technical feasibility; full implementation is not required.

4. Target Users and Use Cases

Target users:

- Municipal cybersecurity and IT operations teams
- Smart-city command-and-control centers
- Electricity, water and other public utility operators
- Traffic, transport and urban-infrastructure authorities

Illustrative scenario: A connected camera suddenly begins transmitting an unusual volume of traffic to an unfamiliar destination. City IoT Shield detects deviation, raises a prioritized alert and recommends that the security team investigate and, if necessary, isolate the device before the issue spreads. Additional use cases include identifying outdated camera firmware, detecting unusual smart-meter communications, discovering unauthorized devices and monitoring high-risk assets across approved city networks.

5. Innovation and Future Vision

City IoT Shield goes beyond a conventional monitoring dashboard by combining asset visibility, anomaly detection, risk prioritization and guided incident response in a single, easy-to-understand platform.

Its modular design could begin with a small pilot and expand as cities adopt autonomous transport, smart grids, AI-enabled services and larger fleets of connected devices.

Its core principles are:

- **Security by design:** Build protection into the platform from the outset rather than adding it later.
- **Zero-trust access:** Continuously verify devices and users before granting access.
- **Early detection:** Identify suspicious behavior before it develops into a wider incident.
- **Resilience:** Support controlled containment, recovery and continuity of essential services.
- **Scalability:** Grow from a limited proof of concept to a broader smart-city deployment.

6. Feasibility and Expected Impact

A focused prototype can be delivered through a four-week plan:

1. **Week 1 — Simulate:** Create representative IoT devices, network activity and sample security events.
2. **Week 2 — Visualize:** Build the device inventory, risk dashboard and monitoring interface.
3. **Week 3 — Detect:** Add anomaly alerts, vulnerability indicators and risk scoring.
4. **Week 4 — Validate:** Test realistic attack scenarios, refine response guidance and demonstrate the end-to-end workflow.

Expected benefits:

- Earlier visibility into emerging cyber threats
- Faster identification of vulnerable or unauthorized devices
- Clearer prioritization of high-impact security risks
- Better protection of sensitive information and operational data
- More resilient and recoverable smart-city services

7. Conclusion

City IoT Shield presents a practical, scalable vision for protecting the connected systems that underpin modern urban life. By unifying device visibility, intelligent threat detection, risk-based prioritisation and guided response, the platform can help city teams act earlier, recover faster and safeguard the services on which citizens rely.

Our Vision: Smarter Cities. Stronger Security. Safer Futures.